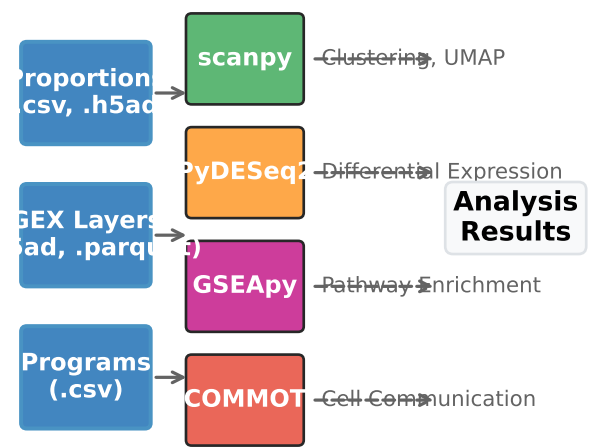
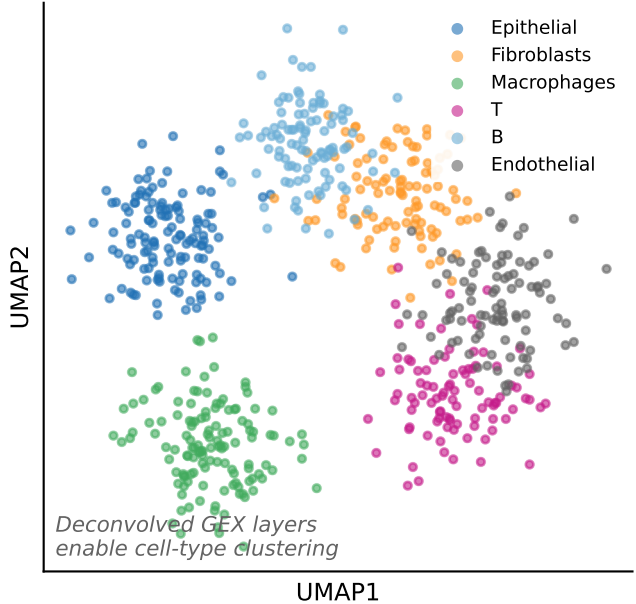


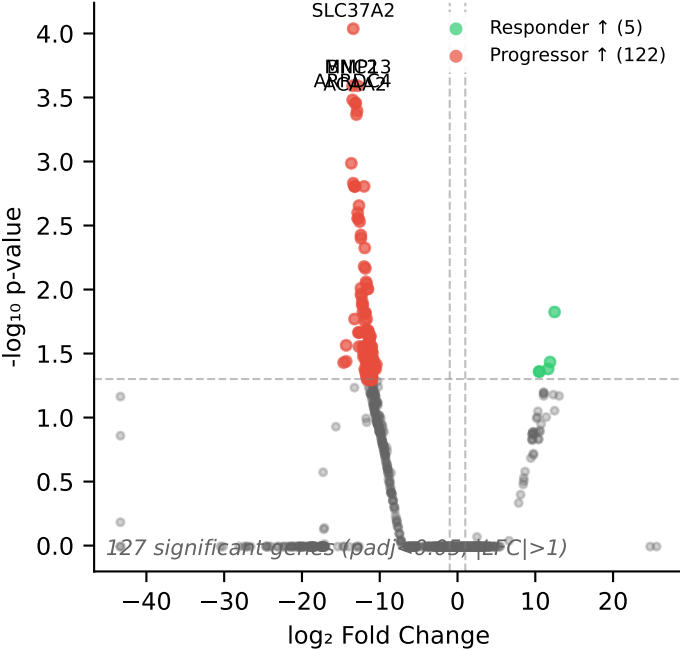
A CITEgeist → Downstream Analysis



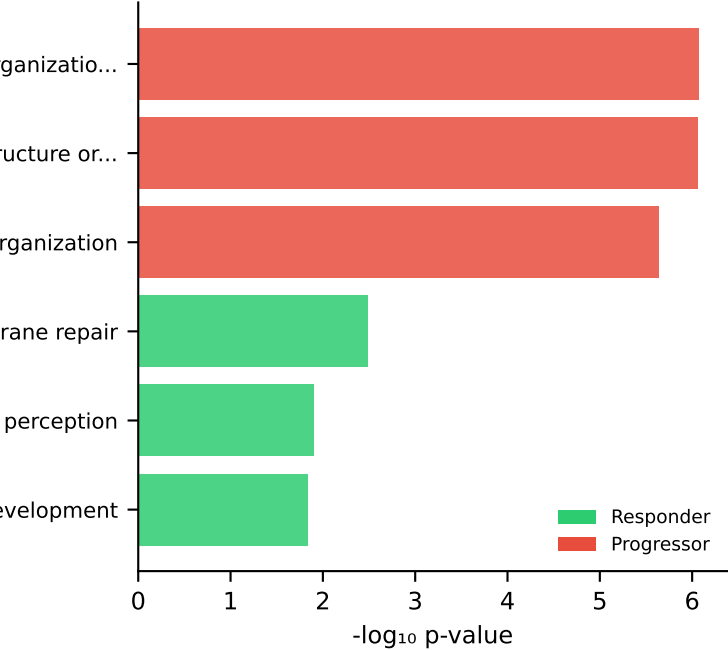
B scanpy: Cluster Visualization



C PyDESeq2: Differential Expression



D GSEAPy: Pathway Enrichment



E Experimental Validation: Midkine Discovery

- **CITEgeist Discovery:** MDK identified as spatially variable in ER+ breast cancer
- **ChIP-seq Validation:** ER binding at MDK locus confirmed, E2-dependent regulation
- **ELISA Confirmation:** Secreted MDK levels correlate with ER signaling
- **Mechanism:** MDK expression modulated by ER-FOXA1 axis

CITEgeist spatial analysis → testable hypothesis → experimental confirmation